

A Noninteger Holder Critical-Line Obstruction for RKHS Sandwiches

Agent lane 7

June 16, 2026

Source and Target

The source is Max Schölpple and Ingo Steinwart, *Which Spaces can be Embedded in Reproducing Kernel Hilbert Spaces?*, arXiv:2312.14711 [1]. The paper proves that the existence of an intermediate reproducing kernel Hilbert space H between two Banach spaces of functions $E \hookrightarrow F$ is equivalent, in the Banach function space setting considered there, to the 2-factorability of the inclusion $E \hookrightarrow F$. It also records the necessary consequence that such an inclusion must be of cotype 2.

For Holder spaces over bounded subsets of $\mathbb{R}^d$ with nonempty interior, the source proves that no RKHS sandwich

$$\text{Hol}^\alpha(X) \hookrightarrow H \hookrightarrow \text{Hol}^\beta(X)$$

can exist when $\alpha - \beta < d/2$. In the higher smoothness Holder-Zygmund/Besov scale it proves positive results when $\alpha - \beta > d/2$, and identifies the critical line $\alpha - \beta = d/2$ as a remaining border case.

these preparations our result reads as follows.

Theorem 12. *Let (X, d) be a metric space and $0 < \beta \leq \alpha \leq 1$. If X is connected and there exists an RKHS H over X with*

$$\text{Höl}_d^\alpha(X) \hookrightarrow H \hookrightarrow \text{Höl}_d^\beta(X), \quad (12)$$

then there exists a constant $C > 0$ such that for all $0 < \delta < \min\{1, \text{diam } X\}$ we have

$$N_d^{\text{pack}}(X, \delta) \leq C\delta^{-2(\alpha-\beta)}. \quad (13)$$

Moreover, if there exists an RKHS H over X such that

$$\text{Höl}_d^\alpha(X) \hookrightarrow H \hookrightarrow \ell_\infty(X), \quad (14)$$

then there exists a constant $C > 0$, such that (13) holds for $\beta := 0$ and all $0 < \delta < \min\{1, \text{diam } X\}$.

Let us consider a bounded $X \subset \mathbb{R}^d$ with non-empty interior. By Theorem 12 and the packing number bound (8) we conclude that there is no RKHS H satisfying (12) if $\alpha - \beta < d/2$. Moreover, there is no RKHS H satisfying (14) if $\alpha < d/2$. Hence, (1) is infeasible for $d > 2$ within the family of Hölder spaces. We observe that the difference of the smoothness between the Hölder spaces in (12) needs to grow proportional to the underlying dimension in order to enable (1).

In Section 4.5, (26) will generalize Hölder spaces over X , allowing arbitrarily large non-integer smoothness. In Theorem 21 we will show that if $\alpha - \beta > d/2$, then there exists an intermediate RKHS H such that we have

$$\text{Höl}^\alpha(X) \hookrightarrow H \hookrightarrow \text{Höl}^\beta(X)$$

and in Theorem 22 we will show that for $\alpha > d/2$ there exists an intermediate RKHS H such that

$$\text{Höl}^\alpha(X) \hookrightarrow H \hookrightarrow C^0(X)$$

holds. Hence, the bounds derived in Theorem 12 are tight up to the border cases $\alpha - \beta = d/2$, respectively $\alpha = d/2$. For general metric spaces (X, d) however, we do not know any condition that implies the existence of an RKHS H satisfying (12) or (14).

The requirement that X is a *bounded* subset of $\mathbb{R}^d$ is crucial to obtain positive answers to (1), as the following

Result

This packet proves a broad negative result on that critical line. We write $\Lambda^s([0, 1]^d)$ for the Holder-Zygmund space $B_{\infty, \infty}^s$; for noninteger $s = m + \theta$, $m \in \mathbb{N}_0$, $0 < \theta < 1$, this is the usual classical Holder space $C^{m, \theta}$, up to equivalence of norms. One concrete equivalent seminorm is given by finite differences: if $r > s$ is an integer and

$$\Delta_h^r f(x) = \sum_{\nu=0}^r (-1)^{r-\nu} \binom{r}{\nu} f(x + \nu h),$$

where only admissible x, h with all points $x, \dots, x + rh$ in $[0, 1]^d$ are used, then

$$\|f\|_{\Lambda^s} \simeq \|f\|_{\infty} + \sup_{0 < |h| \leq 1} |h|^{-s} \|\Delta_h^r f\|_{\infty}.$$

Theorem 1. *Let $d \geq 1$, $\beta > 0$, and*

$$\alpha = \beta + d/2.$$

Assume that $\alpha \notin \mathbb{N}$. Then the inclusion

$$\Lambda^{\alpha}([0, 1]^d) \hookrightarrow \Lambda^{\beta}([0, 1]^d)$$

is not cotype 2. Consequently, there is no reproducing kernel Hilbert space H on $[0, 1]^d$ such that

$$\Lambda^{\alpha}([0, 1]^d) \hookrightarrow H \hookrightarrow \Lambda^{\beta}([0, 1]^d).$$

Since α is noninteger, the domain in Theorem 1 is the ordinary classical Holder space $C^{\alpha}([0, 1]^d)$. The target exponent β may be integer or noninteger; the Holder-Zygmund formulation is used so that integer target endpoints are included in the same statement.

Idea of the Proof

The source's metric packing obstruction tests one scale at a time. At the critical line $\alpha - \beta = d/2$, a single dyadic scale is exactly balanced: there are 2^{jd} available bumps at scale 2^{-j} , and each bump has target norm of size $2^{-j(\alpha-\beta)} = 2^{-jd/2}$. Thus one level contributes a fixed amount to the cotype square-sum.

The extra observation is that one may stack all dyadic levels. If α is noninteger, signed smooth bumps of height $2^{-j\alpha}$ at every level remain uniformly bounded in C^{α} . The lower levels are controlled by their Lipschitz constants, and the higher levels by their small sup norms. But in Λ^{β} the square-sum receives a fixed positive contribution from every level, so the first m levels force a $\sqrt{m}$ lower bound. This contradicts cotype 2, and the source's 2-factorability theorem then excludes an intermediate RKHS.

Multiscale Bump Lemma

Fix a nonzero function $\phi \in C_c^{\infty}((0, 1/4)^d)$. For $j \geq 0$ and $k = (k_1, \dots, k_d) \in \{0, \dots, 2^j - 1\}^d$, set

$$\phi_{j,k}(x) = 2^{-j\alpha} \phi(2^j x - k), \quad x \in [0, 1]^d.$$

The supports of $\{\phi_{j,k}\}_k$ are separated within each fixed level j , and there are 2^{jd} of them.

Lemma 1. *Let $\alpha = n + \theta$ with $n \in \mathbb{N}_0$ and $0 < \theta < 1$. There exists a constant C_α such that for every finite set*

$$A \subset \{(j, k) : j \geq 0, k \in \{0, \dots, 2^j - 1\}^d\}$$

and every choice of signs $\varepsilon_{j,k} \in \{-1, 1\}$,

$$\left\| \sum_{(j,k) \in A} \varepsilon_{j,k} \phi_{j,k} \right\|_{C^{n,\theta}([0,1]^d)} \leq C_\alpha.$$

Consequently the same signed sums are uniformly bounded in $\Lambda^\alpha([0,1]^d)$.

Proof. For a fixed level j , define

$$g_j = \sum_{k:(j,k) \in A} \varepsilon_{j,k} \phi_{j,k}.$$

Let ρ be a multi-index with $|\rho| \leq n$. Since supports at level j are separated and each scaled bump is supported in a cube of side length $2^{-j}/4$,

$$\|D^\rho g_j\|_\infty \leq C_\rho 2^{-j(\alpha-|\rho|)}.$$

For $|\rho| = n$, the same scaling and the separation of supports give the global Lipschitz estimate

$$\text{Lip}(D^\rho g_j) \leq C_\rho 2^{j(1-\theta)}.$$

Indeed, points in the same support are controlled by differentiating one more time, points in different supports are separated by a distance comparable to 2^{-j} , and outside the supports the derivative is zero.

The lower-order C^n part is bounded because

$$\sum_{j \geq 0} 2^{-j(\alpha-|\rho|)} < \infty \quad (|\rho| \leq n).$$

It remains to control the θ -Holder seminorms of the derivatives D^ρ with $|\rho| = n$. Let $x, y \in [0,1]^d$, $x \neq y$, and put $h = |x - y|$. Choose $J \geq 0$ so that

$$2^{-(J+1)} < h \leq 2^{-J},$$

with the evident modification if $h > 1/2$. For the low levels $j \leq J$, the Lipschitz estimate gives

$$\sum_{j \leq J} |D^\rho g_j(x) - D^\rho g_j(y)| \leq Ch \sum_{j \leq J} 2^{j(1-\theta)} \leq C_\theta h^\theta.$$

For the high levels $j > J$, the sup-norm estimate gives

$$\sum_{j > J} |D^\rho g_j(x) - D^\rho g_j(y)| \leq C \sum_{j > J} 2^{-j\theta} \leq C_\theta h^\theta.$$

This proves the uniform $C^{n,\theta}$ bound. The equivalence $C^{n,\theta} = \Lambda^\alpha$ for noninteger α gives the final assertion. $\square$

Lemma 2. *For every $\beta > 0$ there exists $c_\beta > 0$ such that, for all j, k ,*

$$\|\phi_{j,k}\|_{\Lambda^\beta([0,1]^d)} \geq c_\beta 2^{-j(\alpha-\beta)}.$$

Proof. Choose an integer $r > \beta$. Since ϕ is nonzero and compactly supported, its r -th finite differences are not identically zero. Hence there exist $x_0 \in (0, 1/4)^d$, a vector $h_0 \in \mathbb{R}^d$, and a number $\eta > 0$ such that all points $x_0, \dots, x_0 + rh_0$ remain in $(0, 1/4)^d$ and

$$|\Delta_{h_0}^r \phi(x_0)| \geq \eta.$$

For each j, k , put $x_{j,k} = 2^{-j}(k + x_0)$ and $h_j = 2^{-j}h_0$. Then all points used in the finite difference of $\phi_{j,k}$ are admissible, and scaling gives

$$\Delta_{h_j}^r \phi_{j,k}(x_{j,k}) = 2^{-j\alpha} \Delta_{h_0}^r \phi(x_0).$$

Therefore the finite-difference norm of Λ^β yields

$$\|\phi_{j,k}\|_{\Lambda^\beta} \geq c |h_j|^{-\beta} |\Delta_{h_j}^r \phi_{j,k}(x_{j,k})| \geq c_\beta 2^{j\beta} 2^{-j\alpha} = c_\beta 2^{-j(\alpha-\beta)}.$$

□

Proof of Theorem 1

Let

$$A_m = \{(j, k) : 0 \leq j < m, k \in \{0, \dots, 2^j - 1\}^d\}.$$

By Lemma 1, for independent Rademacher signs $\varepsilon_{j,k}$,

$$\mathbb{E} \left\| \sum_{(j,k) \in A_m} \varepsilon_{j,k} \phi_{j,k} \right\|_{\Lambda^\alpha} \leq C_\alpha.$$

On the other hand, Lemma 2 and the critical identity $\alpha - \beta = d/2$ give

$$\begin{aligned} \left(\sum_{(j,k) \in A_m} \|\phi_{j,k}\|_{\Lambda^\beta}^2 \right)^{1/2} &\geq c_\beta \left(\sum_{j=0}^{m-1} 2^{jd} 2^{-2j(\alpha-\beta)} \right)^{1/2} \\ &= c_\beta \left(\sum_{j=0}^{m-1} 1 \right)^{1/2} = c_\beta \sqrt{m}. \end{aligned}$$

Thus the cotype-2 constants of the inclusion

$$\Lambda^\alpha([0, 1]^d) \hookrightarrow \Lambda^\beta([0, 1]^d)$$

are at least $c_\beta C_\alpha^{-1} \sqrt{m}$ for every m , and so the inclusion is not cotype 2.

By the source paper's 2-factorization theorem, an RKHS sandwich

$$\Lambda^\alpha([0, 1]^d) \hookrightarrow H \hookrightarrow \Lambda^\beta([0, 1]^d)$$

would make the inclusion $\Lambda^\alpha([0, 1]^d) \hookrightarrow \Lambda^\beta([0, 1]^d)$ 2-factorable [1, Theorem 6]. By the source paper's type/cotype obstruction, every 2-factorable inclusion is cotype 2 [1, Lemma 5 and Corollary 8]. This contradicts the preceding paragraph and proves the theorem.

Limitations and Remaining Border

The assumption $\alpha \notin \mathbb{N}$ is essential for the present argument: it is exactly what gives the uniform signed multiscale bound in the classical Holder domain norm. The remaining critical cases are the integer-domain smoothness endpoints. They are not a formal afterthought. In dimension one, the endpoint $\alpha = 1$, $\beta = 1/2$ is positive, since

$$C^1([0, 1]) \hookrightarrow H^1([0, 1]) \hookrightarrow C^{1/2}([0, 1])$$

and $H^1([0, 1])$ is a bounded-kernel RKHS. Thus the integer line requires separate tools and is the next genuine frontier.

Verification and Novelty Check

The proof is noncomputational. The verification points are:

1. the signed multiscale bump family is uniformly bounded in $C^\alpha = \Lambda^\alpha$ for noninteger α ;
2. every bump has the stated Λ^β lower bound, including integer β , by finite differences;
3. the critical identity $\alpha - \beta = d/2$ makes every dyadic level contribute a fixed positive amount to the target square-sum;
4. Schölppl and Steinwart’s RKHS theorem converts any sandwich into a 2-factorization, hence into a cotype-2 inclusion.

The run indexes were searched for arXiv id 2312.14711, Holder critical line terminology, $\alpha - \beta = d/2$, cotype, and 2-factorability. The existing hits are the all-dimensional bounded-kernel endpoint packet and a one-dimensional companion Holder-to-Holder packet, not this noninteger critical-line theorem. Bounded web searches on June 16, 2026 for the source title plus “border cases”, for “Holder RKHS alpha-beta=d/2 border cases”, for “Holder-Zygmund RKHS $B_{\infty,\infty}$ border”, and for “ $B_{\infty,\infty}^s$ 2-factorable RKHS Holder critical border” found the source arXiv record but no exact later answer to this subcase.

Human Review Recommendation

Review as a likely valid partial result. The highest-value checks are the finite-difference lower bound in Lemma 2, the global Lipschitz estimate for the separated same-level bumps in Lemma 1, and the precise interpretation of Holder-Zygmund endpoints against the conventions used in the source paper.

References

- [1] Max Schölppl and Ingo Steinwart. *Which Spaces can be Embedded in Reproducing Kernel Hilbert Spaces?* arXiv:2312.14711, 2023.
- [2] Hans Triebel. *Theory of Function Spaces*. Birkhäuser, 1983.